
REQUIREMENT ANALYSIS DOCUMENT

Offline Inventory Management & Billing System

1. Introduction

1.1 Purpose

This document defines the functional and non-functional requirements for an **offline mobile application** designed to manage inventory and generate bills for small businesses. The system ensures smooth operation without internet connectivity.

1.2 Scope

The application will:

- Manage product inventory
- Generate and store invoices
- Track sales transactions
- Work completely offline using a local database

The system is targeted for:

- Small shop owners
 - Retail businesses
 - Local vendors
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1.3 Definitions

Term	Description
Inventory	List of products with quantity and pricing
Invoice	Bill generated for a transaction
SKU	Unique identifier for products
Offline Mode	No internet required

2. Overall Description

2.1 Product Perspective

The system is a **standalone mobile application** that:

- Stores all data locally using SQLite
- Does not depend on servers or APIs
- Ensures quick and reliable access

2.2 Product Functions

The application will provide:

- Product management
- Inventory tracking
- Invoice generation
- Sales history viewing

2.3 User Characteristics

User Type	Description
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Shop Owner	Basic smartphone user, minimal technical knowledge
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User Type	Description
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Staff	Operates billing and inventory updates
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2.4 Constraints

- Must work without internet
 - Limited device storage and performance
 - Simple and intuitive UI required
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3. Functional Requirements

3.1 Product Management

Description

Allows users to manage inventory items.

Functional Requirements

- FR1: User can add a new product
- FR2: User can edit product details
- FR3: User can delete a product
- FR4: User can view product list

Input Fields

- Product Name
 - SKU
 - Quantity
 - Cost Price
 - Selling Price
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3.2 Inventory Management

Description

Tracks available stock levels.

Functional Requirements

- FR5: System updates stock after each sale
 - FR6: System displays current stock levels
 - FR7: System prevents negative stock
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3.3 Billing / Invoice Generation

Description

Generates bills for customer purchases.

Functional Requirements

- FR8: User can create a new invoice
 - FR9: User can add multiple products to invoice
 - FR10: System calculates total amount automatically
 - FR11: System stores invoice details locally
 - FR12: User can view generated invoice
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3.4 Sales Management

Description

Maintains record of all sales.

Functional Requirements

- FR13: System records each transaction
 - FR14: User can view sales history
 - FR15: User can filter sales by date
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3.5 Data Storage (Offline)

Description

Handles local data persistence.

Functional Requirements

- FR16: System stores all data locally using database
 - FR17: Data persists after app restart
 - FR18: No internet required for any operation
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4. Data Requirements

4.1 Entities

Product

- id
 - name
 - SKU
 - quantity
 - cost_price
 - selling_price
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Invoice

- id
 - date
 - total_amount
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Invoice Items

- id
- invoice_id

- product_id
 - quantity
 - price
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5. 🚀 Non-Functional Requirements

5.1 Performance

- App should respond within 2 seconds
 - Should handle at least 1000 products efficiently
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5.2 Usability

- Simple UI for non-technical users
 - Easy navigation
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5.3 Reliability

- Data should not be lost after app closure
 - Proper error handling
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5.4 Security

- Basic data protection within device
 - Prevent accidental deletion
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5.5 Maintainability

- Modular code structure
 - Easy to update features
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6. 📄 System Requirements

Hardware

- Android smartphone
- Minimum 2GB RAM

Software

- Android OS (Version 7.0+)
 - Local DB: SQLite
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7. 🔄 System Workflow

Billing Flow:

Open App → Select/Create Invoice → Add Products → Calculate Total → Save Invoice → Update Inventory

8. ⚠️ Assumptions & Dependencies

- Users have basic smartphone knowledge
 - Device storage is sufficient
 - No dependency on internet
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9. 📌 Summary

This system provides a **lightweight, offline-first solution** for inventory and billing. It ensures:

- Simplicity
 - Reliability
 - Accessibility for small businesses
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